

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Easy

Time allowed: 30

Score: /17

Percentage: /100

Question 1a

This question is about different polymers.

Polymer **X** can be formed from an alkene. The structure of polymer **X** is shown in Fig. 1.1.

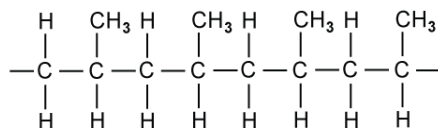


Fig. 1.1

State the type of polymerisation involved in the formation of polymer **X**.

[1 mark]

Question 1b

i)

Draw the monomer used to form polymer **X**.

[1]

ii)

Name polymer **X**.

[1]

[2 marks]

Question 1c

Suggest **three** reasons why the disposal of polymer **X** would prove to be a challenge.

[3 marks]

Question 2a

Butene, C_4H_8 , can exist as three isomers.

Isomer **A** is a position isomer of isomers **B** and **C** and does not show geometric isomerism.

Isomers **B** and **C** are geometric isomers of each other.

Isomer **B** is the cis isomer.

Isomer **C** is the trans isomer.

i)

Draw the displayed formulae of the three isomers, **A**, **B** and **C**, of butene, C_4H_8 .

[3]

ii)

Name all three isomers, **A**, **B** and **C**, of butene, C_4H_8 .

[3]

[6 marks]

Question 2b

Draw the displayed formula of the addition polymer formed by isomer **A**.

[1 mark]

Question 2c

A student was asked to draw the displayed formula of the addition polymer of isomers **B** and **C**. They suggested that there were two structures as shown in Fig. 2.1.

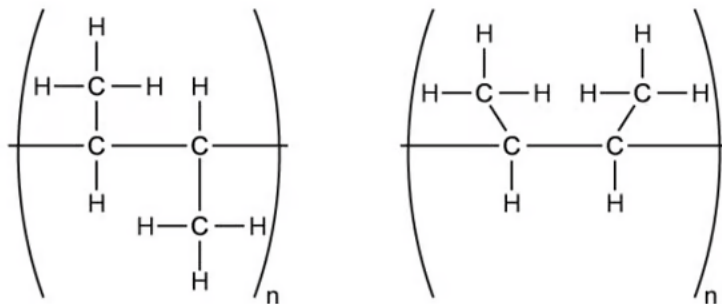


Fig. 2.1

State why the student's two suggested structures are the same addition polymer.

[1 mark]

Question 3a

A molecule of chloroethene is shown in Fig. 3.1.

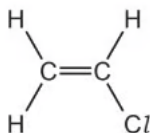


Fig. 3.1

Chloroethene can be polymerised to form poly(chloroethene).

State the alternative name for poly(chloroethene).

[1 mark]

Question 3b

Draw a diagram of the structure of poly(chloroethene) including **three** repeat units.

[1 mark]

Question 3c

State a problem caused by the combustion of poly(chloroethene) as a method of disposal and suggest how the effects of this problem can be reduced.

[1 mark]

Model Answers: Easy

1a

a) The name of the type of polymerisation is:

- Addition polymerisation; [1 mark]

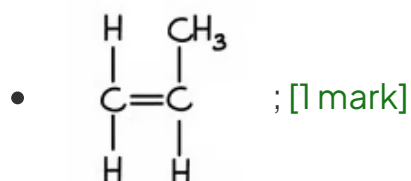
[Total: 1 mark]

- Addition polymerisation is the reaction in which many monomers containing at least one C=C double bond form long chains of polymer as the only product

1b

b)

i) The monomer to form polymer X is:

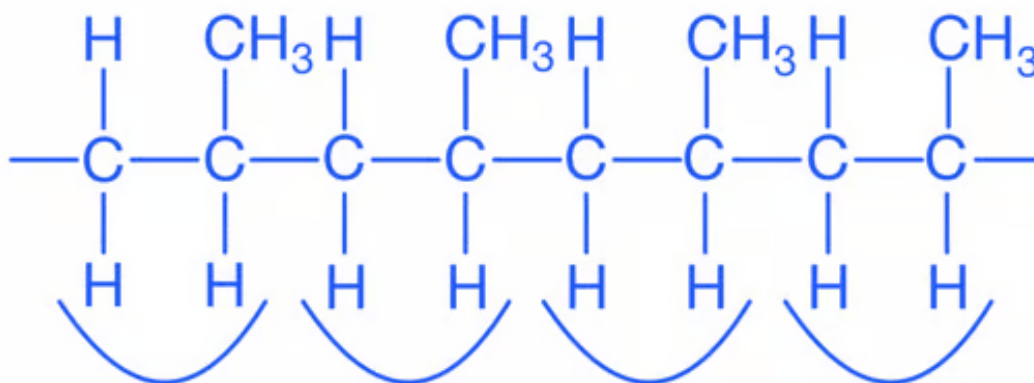


ii) The name of polymer X is:

- Poly(propene); [1 mark]

[Total: 2 marks]

- **Remember:** A repeating unit only has 2 carbons in the polymer chain
- In this case, the repeat unit is two carbon atoms where one of them has an attached CH₃ group



- To draw the monomer, put a C=C double bond between the two carbon atoms in the repeating unit of the polymer chain and remove the bond attached to the carbon atom on the other side

1c

b) **Three** reasons that disposing of polymers / plastics is a problem are:

Any **three** of the following;

- They are unreactive; [1 mark]
- They are resistant to chemical attack; [1 mark]
- They are non-biodegradable

OR

They can take hundreds of years to decompose; [1 mark]

- They form harmful / toxic chemicals when burnt

OR

They release carbon dioxide / pollutant gases when burnt; [1 mark]

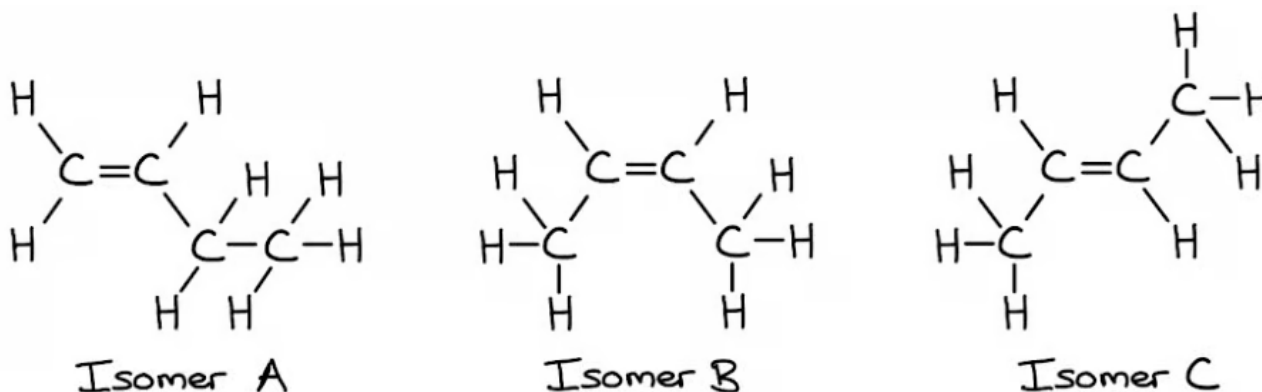
[Total: 3 marks]

- You should already know the main methods of disposal for polymers / plastics and the associated advantages and, more importantly, disadvantages:
 - Burn - air pollution
 - Bury - visual / air pollution, uses up land that could be better used
- You may also be aware of the high time demand in recycling polymers / plastics
- This question asks why they are such a challenge without covering these methods directly

2a

a)

i) The displayed formulae of the three isomers, **A**, **B** and **C**, of butene, C_4H_8 are:



- Correct structure for isomer **A**; [1 mark]
- Correct structure for isomer **B**; [1 mark]
- Correct structure for isomer **C**; [1 mark]

ii) The names of the three isomers, **A**, **B** and **C**, of butene, C_4H_8 are:

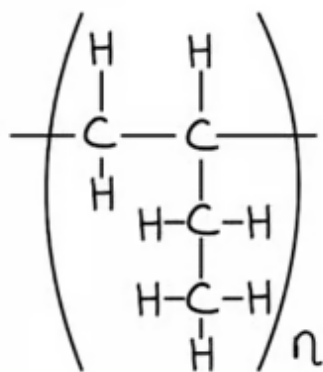
- Isomer **A** = but-1-ene; [1 mark]
- Isomer **B** = *cis*-but-2-ene; [1 mark]
- Isomer **C** = *trans*-but-2-ene; [1 mark]

[Total: 6 marks]

- When drawing isomer structures of an alkene, you should start with the $C=C$ bond
- Isomers **B** and **C** are geometric isomers, so they must have methyl / CH_3 groups attached at the “ends” of the $C=C$
 - You then have to use the *cis* and *trans* information about isomers **B** and **C** to deduce which isomer is which
- Isomer **A** is a position isomer of **B** and **C** and it doesn't show geometric isomerism
 - The only way that this is possible is for isomer **A** to have one ethyl group, making it but-1-ene

2b

c) The displayed formula of the addition polymer formed by isomer **A** is:



; [1 mark]

[Total: 1 mark]

- To draw an addition polymer
 - Draw the monomer
 - Replace the C=C with a C-C bond
 - Put continuation bonds on either side of the C-C bond
 - You can also add brackets and an n but this is not essential, just good practice

2c

d) The student's two suggested structures are the same addition polymer because:

- There is free rotation around the C-C

OR

The C-C bond can freely move / rotate

OR

There is no longer a C=C to restrict / stop rotation; [1 mark]

[Total: 1 mark]

- **Remember:** Single bonds have free rotation
- This means that for addition polymers, one pair of isomers can sometimes give only one possible addition polymer

3a

a) The alternative name for poly(chloroethene) is:

- Poly(vinyl chloride) / PVC; [1 mark]

2c

d) The student's two suggested structures are the same addition polymer because:

- There is free rotation around the C-C

OR

The C-C bond can freely move / rotate

OR

There is no longer a C=C to restrict / stop rotation; [1 mark]

[Total: 1 mark]

- **Remember:** Single bonds have free rotation
- This means that for addition polymers, one pair of isomers can sometimes give only one possible addition polymer

3a

a) The alternative name for poly(chloroethene) is:

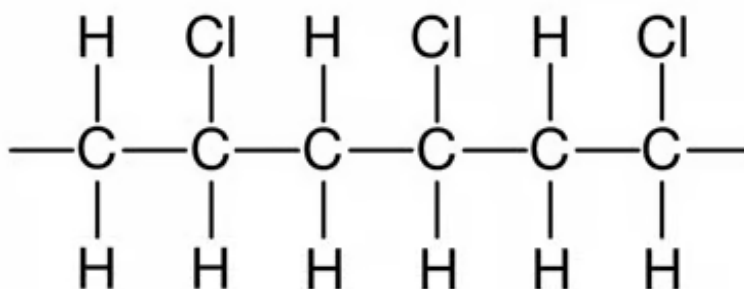
- Poly(vinyl chloride) / PVC; [1 mark]

[Total: 1 mark]

- You are expected to know the alternative and IUPAC name of poly(chloroethene) and to be able to draw its structure

3b

b) A diagram of the structure of poly(chloroethene) including **three** repeat units is:



- Correct structure of repeat unit; [1 mark]
- Three repeat units drawn **AND** continuations bonds at either end; [1 mark]

[Total: 2 marks]

- From the monomer:
 - Replace the C=C bond with a C-C bond
 - Put continuation bonds on either side of the C-C bond
 - Join 3 of these repeat units together

3c

c) A problem caused by the combustion of poly(chloroethene) as a method of disposal and a possible way to reduce its effects are:

Any **one** of the following problem and suggestion to reduce effect:

- Problem: Acidic hydrogen chloride gas is given off; [1 mark]
- Suggestion to reduce effect: Neutralise (acidic gases / hydrogen chloride) before releasing; [1 mark]

OR

- Problem: Dioxins / toxic compounds are released; [1 mark]
- Suggestion to reduce effect: Use very high temperature during combustion (to break toxins down); [1 mark]

[Total: 2 marks]

- The high chlorine content in poly(chloroethene) is problematic when it comes to its disposal
 - Recycling is a much more sustainable option
- You should be aware that hydrogen chloride gas is given off during combustion of poly(chloroethene)
- To reduce the effects of acidic hydrogen chloride, you need to think of a way to reduce the amount being released into the atmosphere
- As it is acidic, it can be neutralised by a base prior to the waste gases being released